Fundamentals of Operations Management

How Systems Are Designed, Run, and Improved

REVIEW

Quick Recap From Last Class

What is operations management?

The discipline of designing, running, and continuously improving systems that deliver value to customers.

Where does operations start and end?

Operations begins with system design and continues indefinitely—there is no end point as long as the system serves customers.

Why is it continuous?

Because customer needs evolve, systems degrade, and competitive advantages require constant adaptation and improvement.

> Operations management ensures consistent value delivery, not just system creation.

What Operations Management Actually Manages

Operations management orchestrates the critical elements that transform strategy into execution:

- Processes and workflows
- Resources and assets
- Flow of work through systems
- Capacity and throughput
- Quality standards and consistency
- Cost efficiency and optimization
- Reliability and uptime

Operations is the bridge between strategy and execution.

Core Elements of Operations Management

The fundamentals of operations can be organized into five essential pillars that will appear repeatedly throughout your profession in any industry:

Process

The sequence of activities that transforms inputs into outputs

Resources

The people, technology, time, and capital required to operate

Capacity

The maximum throughput a system can handle effectively

Quality

Consistency in meeting customer expectations and standards

Continuous Improvement

Ongoing optimization and adaptation of operations

Process: The Heart of Operations

Every operation follows this universal model, defining what happens, in what order, and by whom.

Software Development Example

Input: Code and requirements

Process: Build, test, deploy

Output: Running software service

Industry	Input	Output
Software	Code, data, requirements	Running application
IT Services	Client requirements	Client solution
Hardware	Raw materials	Physical product

Operations Perspective: Software Industry

Operational Focus

- Continuous deployment
- System uptime and availability
- Scalability under load

Key Operations Activities

- Version control management
- Automated testing and deployment
- Real-time monitoring and incident response

Failure Impact

- System downtime
- Revenue loss
- User dissatisfaction and churn

> Software operations never stop—even at midnight, on holidays.

Operations Perspective: Service-Based IT

Operational Focus

- Meeting client commitments
- Service Level Agreements (SLAs)
- On-time delivery of solutions

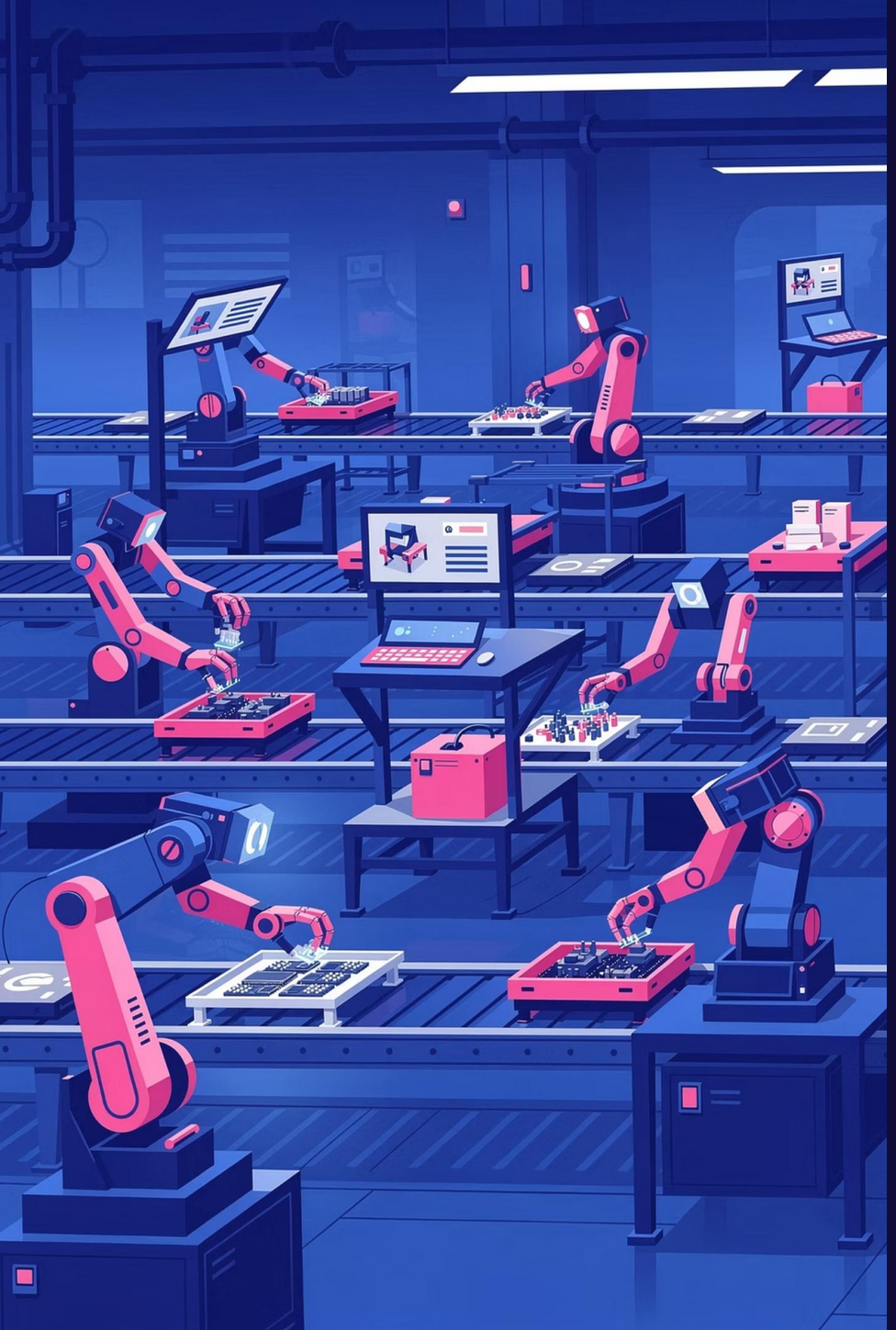
Key Operations Activities

- Resource allocation (engineers per project)
- Scheduling and workload balancing
- Incident and escalation management

Critical Metric

Client satisfaction scores directly reflect operational excellence. In IT services, delays and quality issues are immediately visible to clients.

In IT services, failure is visible to the client immediately.



Operations Perspective: Hardware Manufacturing

Operational Focus

- Process stability
- Quality consistency
- Cost efficiency

Key Operations Activities

- Production planning
- Quality control checkpoints
- Inventory and supply coordination

Failure Impact

- Product recalls
- Massive financial loss
- Brand damage

> Software bugs can be patched. Hardware defects are expensive.

Resources in Operations

Operations management must optimize limited resources across multiple dimensions:

Human Resources
Engineers, operators, specialists with specific skills

Technology
Servers, tools, cloud infrastructure, software licenses

Time
Deadlines, schedules, response windows

Money
Budget constraints, operational costs

Even the best system fails if resources are poorly managed.

Capacity Management

Capacity represents the maximum output a system can handle effectively. Understanding and managing capacity prevents both system failure and resource waste.

1

Software Industry

Users per server, requests per second, API throughput, server load

2

IT Services

Engineer hours available, team size, concurrent projects

3

Hardware Manufacturing

Units produced per day, assembly line speed

The Capacity Balance

Demand > Capacity = System failure

Capacity >> Demand = Wasted cost

Effective capacity management means matching resources to actual demand with appropriate buffer for variability. Mismanagement causes either operational failure or inefficient resource utilization.

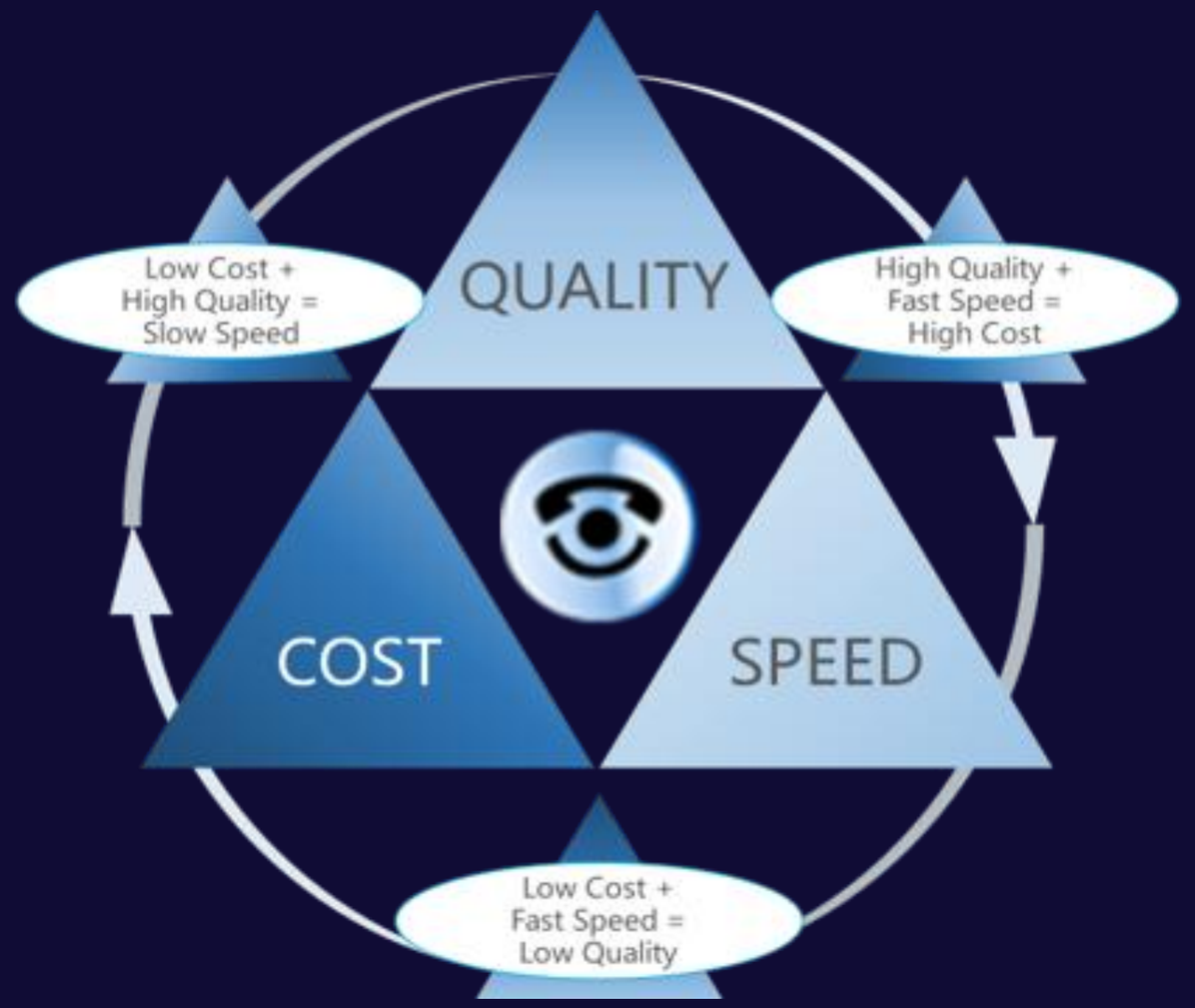
Quality in Operations

Quality in operations means doing the right thing consistently and meeting customer expectations every time.

Industry	Quality Means
Software	Bug-free code, fast response times, reliable performance, high availability
IT Services	SLA compliance, quick response times, meeting deliverables on schedule
Hardware	Defect-free physical products, durability, meeting specifications

> Quality is defined by the customer, not the developer.

Cost, Speed & Reliability: The Trade-off Triangle



Operations must constantly balance these three competing priorities. Improving one dimension often requires compromises in the others.

Industry Priorities

- **Software:** Prioritize reliability and uptime
- **IT Services:** Prioritize delivery speed and SLA compliance
- **Hardware:** Prioritize cost efficiency and quality

⚠ CRITICAL

Why Operations Fail

Understanding failure modes is essential for building resilient operations:

Poor Process Design

Unclear workflows, missing steps, inefficient sequences that create bottlenecks and confusion

Ignoring Capacity Limits

Pushing systems beyond their maximum throughput without planning for scalability

Weak Coordination

Teams working in silos without communication, leading to misalignment and duplicated effort

No Monitoring or Feedback

Flying blind without metrics, unable to detect problems before they become crises

Operations failure ≠ coding failure. Most operational failures stem from design and planning issues, not technical implementation.

Operations Management in Software Development

For software engineers, operations management extends far beyond writing code:

01

Version Control

Managing code repositories, branches, and collaborative development workflows

02

Testing

Automated unit, integration, and end-to-end testing to ensure quality

03

Deployment

CI/CD pipelines, release management, rollback procedures

04

Monitoring

Real-time system health checks, performance metrics, alerting systems

05

Incident Response

Rapid problem detection, diagnosis, resolution, and post-mortem analysis

> "It works on my machine" is not an operations success.

Operations vs Project Work

Understanding the distinction between projects and operations is fundamental to managing both effectively:

Project

- Temporary with defined start and end
- Ends at delivery or launch
- Feature-focused and innovation-driven
- Creates new systems and capabilities
- Success = meeting objectives on time

Operations

- Continuous and ongoing
- Never ends as long as system serves users
- Performance-focused and reliability-driven
- Keeps systems alive and functioning
- Success = consistent value delivery

Projects create value. Operations protect value.

INTERACTIVE

Mini Case: Think–Pair–Share

Duration: 3–4 minutes

Scenario

An application works perfectly during testing with a small user base. However, within hours of public release, the system crashes repeatedly and becomes unavailable to users.

1. **Is this a development issue or operations issue?**

   Consider what went wrong and when

2. **Which fundamentals were ignored?**

   Think about the five pillars of operations

3. **What should have been planned earlier?**

   What steps could have prevented this failure?

Operations Management & Engineers

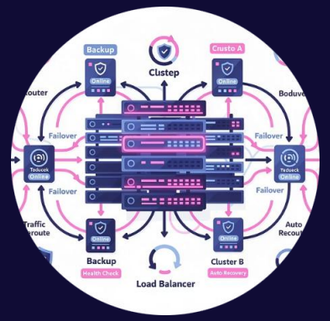
For computer science engineers, mastering operations management means developing a comprehensive systems perspective:

Thinking Beyond Code

Understanding the full system lifecycle from design through deployment and maintenance

Designing for Scale

Building systems that can grow and adapt to increasing demand



Planning for Failure

Anticipating what can go wrong and building resilience

Understanding Business Impact

Recognizing how technical decisions affect revenue, users, and reputation

KEY TAKEAWAYS

Summary: What You Must Remember

Definition

Operations management = designing + running + improving systems

Core Fundamentals

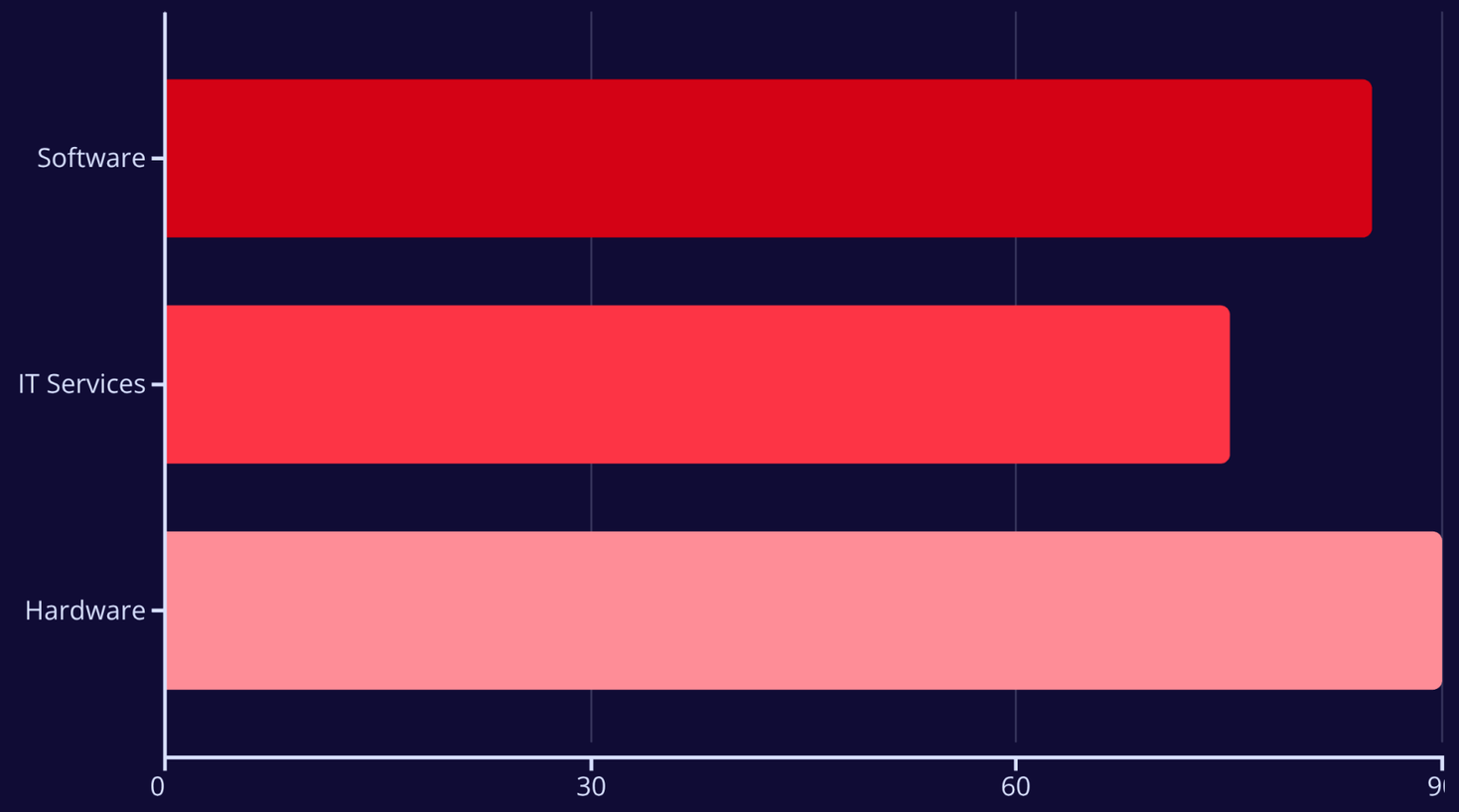
Process, Resources, Capacity, Quality, Continuous Improvement

Continuous Nature

Operations is ongoing, not one-time; it never stops

Operations Fundamentals Apply to All Industries

While the core principles remain constant, implementation varies significantly across sectors:

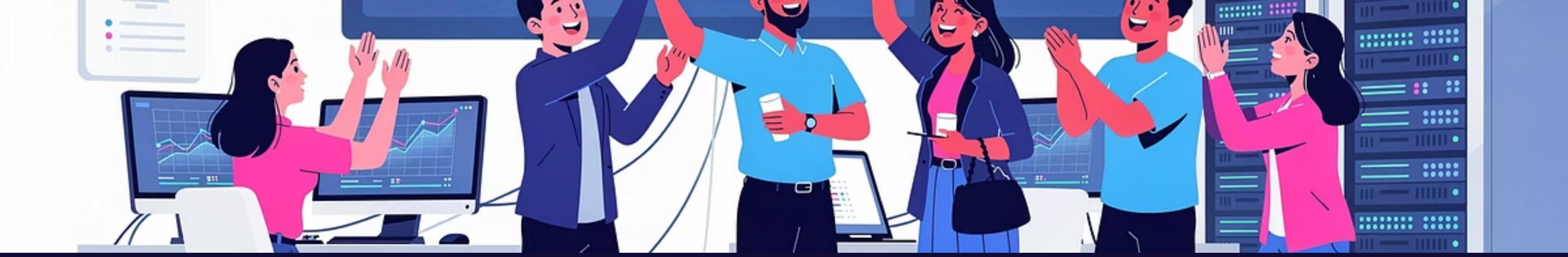


Industry-Specific Considerations

Software: Emphasizes continuous deployment, scalability, and real-time monitoring

IT Services: Focuses on resource allocation, SLA compliance, and client satisfaction

Hardware Manufacturing: Prioritizes process stability, quality control, and cost efficiency



Strong Operations = Competitive Advantage

Organizations that excel at operations management consistently outperform competitors. As you prepare for your careers, remember that technical skills are essential—but operational excellence is what separates good systems from great ones.

3x

Revenue Growth

Companies with strong operations grow faster

50%

Cost Reduction

Efficient operations significantly lower costs

99.9%

Reliability Target

Industry standard for system uptime

> Your ability to design, run, and improve systems will define your impact as an engineer.

Thank you!